HPE JUNIPER networking

Data sheet



SRX2300 firewall data sheet

HPE

Product description

The Juniper Networks SRX2300 firewall is a high performance, next-generation firewall (NGFW) designed to provide reliable network protection for your enterprise campus edge and data center edge. It also supports roaming, SD-WAN large branch, and SD-WAN secure hub use cases. Combining carrier-grade routing with state-of-the-art switching, this platform delivers robust security, effective threat detection, and comprehensive automation and mitigation capabilities.



Figure 1. Juniper SRX Series firewalls have achieved the highest scores in security effectiveness by CyberRatings and NetSecOpen

The SRX2300 firewall delivers NGFW features that support the changing needs of cloud-enabled enterprise networks and data centers. Whether rolling out new services within an enterprise campus, connecting to the cloud seamlessly, complying with industry standards, or achieving operational efficiency, the SRX2300 empowers organizations to operationalize zero trust principles at scale while realizing their business objectives. The SRX2300 protects critical corporate assets with intrusion prevention system (IPS), follow-the-user and follow-the-application access policies, and HPE's AI-Predictive Threat Prevention. Furthermore, the SRX2300 works with Juniper cloud security solutions to secure hybrid cloud environments with networkwide visibility and control, providing consistently secure on-premises and cloud environments.

As network architectures become more distributed and decentralized, Juniper Networks SRX Series firewalls ensure seamless integration with other Hewlett Packard Enterprise and third-party networking platforms. At the same time, the NGFWs facilitate architectural transformation, taking organizations from on-premises to hybrid cloud environments seamlessly and cost-effectively. SRX Series firewalls are the first to implement industry-standard Ethernet VPN (EVPN) Type 5 and Virtual Extensible LAN (VXLAN) protocols within data center environments, enabling the SRX2300 to act as a secure, fabric-aware leaf in the data center spine-leaf architecture.

The SRX2300 participates in the industry-first Connected Security Distributed Services Architecture, enabling organizations to scale horizontally and elastically and simplify the operational management of large-scale firewall networks. With this architecture, several SRX2300 platforms can work together as a single large logical firewall to provide higher performance and scale security.

The SRX2300 is powered by HPE's industry-leading Junos operating system that underpins and helps secure the world's largest mission-critical enterprise and service provider networks. It is managed by Juniper Security Director Cloud, Juniper's unified management experience that connects the organization's current deployments with future architectural rollouts. Security Director Cloud uses a single policy framework, enabling consistent security policies across any environment and expanding zero trust to all parts of the network—from the edge into the data center. This provides unbroken visibility, policy configuration, administration, and collective threat intelligence all in one place.

Product overview

The firewall's role must expand as data centers evolve from traditional architectures to distributed ones. Rather than being a perimeter technology, firewalls need to be part of a security fabric woven throughout the network. A security fabric ensures that security is maintained at every point of connection. Juniper's Connected Security Distributed Services Architecture, managed by Juniper Security Director Cloud, offers a high performance, scalable, and easy-to-manage firewall solution to secure today's distributed data centers. The Juniper Networks SRX2300 firewall is integral to this new architecture and empowers organizations to operationalize security across their networks. This 1U, power-efficient firewall features built-in zero trust, Ethernet VPN-Virtual Extensible LAN (EVPN-VXLAN) fabric integration, and AI-Predictive Threat Prevention to secure your network.

Architecture and key components

The SRX2300 hardware and software architecture provides cost-effective security in a compact, scalable 1U form factor. Purpose built to protect network environments, the SRX2300 incorporates multiple security services and networking functions on top of Junos OS, providing highly customizable threat protection, automation, and integration capabilities. Best-in-class advanced security capabilities on the SRX2300 are offered in the data center, enterprise campus, and regional headquarters deployments with IMIX traffic patterns.

Built-In zero trust

To increase trust and streamline operations, the SRX2300 features several built-in zero trust device capabilities, including an embedded Trusted Platform Module (TPM) 2.0 and cryptographically signed device ID. The SRX2300 supports RFC-compliant secure zero touch provisioning (sZTP) to efficiently, expediently, and remotely deploy products in your network. Additionally, the SRX2300 supports MACsec at wire speed, ensuring data integrity and confidentiality.

Connected Security Distributed Services Architecture

The SRX2300 firewall is part of HPE Juniper Networking Connected Security Distributed Services Architecture, revolutionizing data center security. With HPE Juniper Networking Connected Security Distributed Services Architecture, firewall performance can scale horizontally by interconnecting traffic forwarding and security services across multiple locations. The HPE Juniper Networking solution also provides automated failover and backup nodes for forwarding and inspection components. In addition to redundancy and load balancing, HPE Juniper Networking Connected Security Distributed Services Architecture simplifies how large-scale data center firewall networks are managed and operated. Regardless of how many additional firewall engines across the various form factors (physical, virtual, containerized) are added, they can all be managed as one logical unit. This centralized management eliminates the complexity that has been an unintended consequence of a traditional scale-out approach.

Features and benefits

Business Requirement	Feature/Solution	SRX2300 advantages
High performance	Hardware-accelerated encryption/decryption	— Offloads CPU-intensive encryption/decryption tasks — Improves performance for SSL and IPSec
High-quality end user experience	Application visibility and control	— Updates application continuously and decodes custom applications — Controls and prioritizes traffic based on application and user role — Inspects and detects applications inside SSL-encrypted traffic, including Web and SaaS
Advanced threat protection	NGFW Services: IPS, antivirus, antispam, Web filtering Juniper Advanced Threat Prevention Cloud: sandboxing, Encrypted Traffic Insights, SecIntel threat intelligence feeds	— Prevents exploits with 99.9% effectiveness[1]; signatures update in real time — Protects against known malware and malicious Web and DNS traffic — Sandboxing for unknown malware across multiple OS types, including iOS, Windows, Android™, and CentOS — Delivers threat intelligence in an open platform to accommodate third-party and custom threat feeds — Detects threats hidden inside encrypted traffic without decrypting

[1] Exploit block rate results tested by CyberRatings' 2023 Enterprise Firewall test report

Business requirement	Feature/Solution	SRX2300 advantages
Zero-day protection	HPE Juniper Networking AI-Predictive Threat Prevention	— Predicts and prevents malware at line rate by using AI to identify threats from packet snippets effectively — Eliminates patient-zero infections — Auto-generates protective signatures that remain active for the full attack lifecycle, keeping the network safe from subsequent attacks
Secure data transactions	Juniper Secure Connect: IPSec VPN, remote access/ SSL VPN	— Provides high performance IPSec VPN with dedicated crypto engine — Offers diverse VPN options for various network designs, including remote access and dynamic site-to-site communications — Simplifies large VPN deployments with auto-VPN — Includes hardware-based crypto acceleration — Secure and flexible remote access SSL VPN
Advanced networking services	Routing, secure wire	— Supports carrier-class advanced routing and quality of service (QoS)
Security embedded into the data center fabric	EVPN-VXLAN (EVPN Type 5 route)	— Enhances tunnel inspection for VXLAN encapsulated traffic with Layer 4-7 security services — Eases operations with Type 5 support through BGP — Does not require decapsulation for EVPN-VXLAN traffic
Reliability	Chassis cluster, MNHA, redundant power supplies	— Provides stateful configuration and session state synchronization — Supports active/active and active/backup deployment scenarios — Offers highly available hardware with redundant power supply unit (PSU) and fans
Easy to manage and scale	Juniper Security Director Cloud, on-box GUI	— Provides centralized management via HPE Juniper Networking unified management experience, including zero touch provisioning (ZTP), unbroken visibility, intelligent rule placement, and simplified policy configuration and automation — Supports Network Address Translation (NAT) and automated IPSec VPN deployments via wizards — Supports on-box GUI
Built-in zero trust capabilities	DevID with TPM 2.0 Module	— Verifies the device's trust posture easily — Provides cryptographically signed device ID that supports RFC-compliant sZTP for hardware and software attestation — Mitigates the risks of supply chain attacks
Low TCO	Junos OS	— Integrates routing and security capabilities into a single device — Reduces OpEx with Junos OS automation capabilities — Automates integration with other devices running Junos OS, such as Juniper MX, PTX, and ACX routers; EX and QFX switches; and Cloud-Native Contrail Networking (CN2)

Software specifications

Firewall Services

— Stateful firewall services
— Zone-based firewall
— Screens and distributed denial of service (DDoS) protection
— Protection from protocol and traffic anomalies
— Unified Access Control (UAC)
— Integration with Juniper Mist Access Assurance

Carrier-Grade Network Address Translation (CGNAT)

— Carrier-grade Network Address Translation (Large-scale NAT)
— IPv4 and IPv6 address translation NAT44, NAPT44, NAT66, NAPT66, NAT64, NAT46
— Static and dynamic 1-1 translation
— Source NAT with Port Address Translation (PAT)
— Destination NAT with Port Address Translation (PAT)
— Persistent NAT (EIM/EIF)
— Port Block Allocation (PBA)
— Deterministic NAT (DetNAT)
— Port overload
— Twice-NAT44
— DS-lite and Port Control Protocol (PCP)

VPN features

— Tunnels: Site-to-site, hub and spoke, dynamic endpoint, AutoVPN, ADVPN, Group VPN (IPv4/ IPv6/ Dual Stack)
— Juniper Secure Connect: Remote access IPSec/ SSL VPN
— Configuration payload: Yes
— IKE encryption algorithms: Prime, 3DES-CBC, AEC-CBC, AES-GCM, Suite B
— Authentication: Pre-shared key and public key infrastructure (PKI) (X.509)
— IPSec: Authentication Header (AH) / Encapsulating Security Payload (ESP) protocol
— IPSec authentication algorithms: hmac-md5, hmac-sha-196, hmac-sha-256
— IPSec encryption algorithms: Prime, DES-CBC, 3DES-CBC, AEC-CBC, AES-GCM, Suite B
— Perfect forward secrecy, anti-replay
— Internet Key Exchange: IKEv1, IKEv2
— Monitoring: Standard-based dead peer detection (DPD) support, VPN monitoring
— VPNs GRE, IP-in-IP, and MPLS

High availability features

— Virtual Router Redundancy Protocol (VRRP)–IPv4 and IPv6
— Stateful high availability: Dual box clustering
  • Active/passive
  • Active/active
  • Configuration synchronization
  • Firewall session synchronization
  • Device/link detection
  • In-Service Software Upgrade (ISSU)
  • IP monitoring with route and interface failover
  • BFD monitoring
— Chassis cluster HA and Multinode HA (MNHA)

Application Security Services (offered as advanced security subscription license)

— Application visibility and control
— Application QoS
— Advanced/application policy-based routing (APBR)
— Application Quality of Experience (AppQoE)
— Application-based multipath routing
— User-based firewall

Threat Defense and Intelligence Services (offered as an advanced security subscription license)

— Intrusion prevention system
— AI-Predictive Threat Prevention
— Antivirus
— Antispam
— Category/reputation-based URL filtering
— SSL proxy/inspection

— Protection from botnets (command and control)

— Adaptive enforcement based on GeoIP

— Juniper Advanced Threat Prevention, a cloud-based SaaS offering to detect and block zero-day attacks

— Adaptive Threat Profiling

— Encrypted Traffic Insights

— SecIntel threat intelligence

— Juniper ATP virtual appliance, a distributed, on-premises advanced threat prevention solution to detect and block zero-day attacks

Routing protocols

— IPv4, IPv6, static routes, RIP v1/v2

— OSPF/OSPF v3

— BGP with route reflector

— IS-IS

— Multicast: Internet Group Management Protocol (IGMP) v1/v2, Protocol Independent Multicast (PIM) sparse mode (SM)/source-specific multicast (SSM), Session Description Protocol (SDP), Distance Vector Multicast Routing Protocol (DVMRP), Multicast Source Discovery Protocol (MSDP), reverse path forwarding (RPF)

— Encapsulation: VLAN, Point-to-Point Protocol over Ethernet (PPPoE)

— Virtual routers

— EVPN-VXLAN (EVPN Type 5 route)

— Policy-based routing, source-based routing

— Equal-cost multipath (ECMP)

QoS features

— Support for 802.1p, DiffServ code point (DSCP), EXP

— Classification based on VLAN, data-link connection identifier (DLCI), interface, bundles, or multifield filters

— Marking, policing, and shaping

— Classification and scheduling

— Weighted random early detection (WRED)

— Guaranteed and maximum bandwidth

— Ingress traffic policing

— Virtual channels

Network services

— Dynamic Host Configuration Protocol (DHCP) client/server/relay

— Domain Name System (DNS) proxy, dynamic DNS (DDNS)

— HPE Juniper Networking real-time performance monitoring (RPM) and IP monitoring

— Juniper flow monitoring (J-Flow)

Advanced Routing Services

— MPLS (RSVP, LDP)

— Circuit cross-connect (CCC), translational cross-connect (TCC)

— L2/L2 MPLS VPN, pseudo-wires

— Virtual private LAN service (VPLS), next-generation multicast

— VPN (NG-MVPN)

— MPLS traffic engineering and MPLS fast re-route

Management, automation, logging, and reporting

— SSH, Telnet, SNMP-MIBs, Traps

— Smart image download

— Juniper CLI, Web UI, NetCONF, XML APIs, RMON

— Juniper Networks Security Director Cloud

— Python

— Junos OS events, commit and OP scripts

— Application and bandwidth usage reporting

— Debug and troubleshooting tools

Hardware specifications

Table 3. SRX2300 Hardware Specifications

Specifications	SRX2300
Connectivity	
Onboard ports	8 x 1GbE/2.5GbE/ 5GbE/10GbE BASE-T
Onboard small form-factor pluggable plus (SFP+) transceiver ports	8 x 1GbE/10GbE SFP+ 4 x 1GbE/10GbE/25GbE SFP28 2 x 40GbE/100GbE QSFP28
Out-of-Band (OOB) management ports	1 x 1GbE (RJ-45)
Dedicated high availability (HA) ports	2 x 1GbE SFP
Console	1 (RJ-45)
USB 3.0 ports (Type A)	1
Storage	
Storage (SSD)	1 x 120 GB (primary), 1 x 120 GB (secondary)
Dimensions and Power	
Form factor	1U
Size (W x H x D)	17.28 x 1.74 x 18.20 in (43.89 x 4.42 x 46.23 cm)
Weight (device and PSU)	Chassis with two AC power supplies: 19 lb (8.6 kg) Chassis with two DC power supplies: 19.3 lb (8.8 kg) Chassis with package for shipping: 35.6 lb (16.2 kg)
Redundant PSU	1+1
Power supply	2 x 450W AC PSU redundant 2 x 650W DC PSU redundant

Specifications	SRX2300
Average heat dissipation	1 x DC PSU (40V): 653.4 BTU/h 2 x DC PSU (40V): 737 BTU/h 1 x AC PSU (110V): 682 BTU/h 1 x AC PSU (230V): 662 BTU/h 2 x AC PSU (110V): 703 BTU/h 2 x AC PSU (230V): 682 BTU/h
Environment and Regulatory Compliance	
Airflow/cooling	Front to back
Operating temperature	32°F to 104°F (0°C to 40°C at 6000 ft altitude)
Operating humidity	5% to 90% non-condensing
Meantime between failures (MTBF)	Over 100,000 hours (12 years)
FCC classification	Class A
RoHS compliance	RoHS 6
Performance and Scale	
Firewall throughput[2] (IMIX)	28 Gbps
Firewall throughput[2] (1518B)	39 Gbps
IPSec VPN throughput[2] (IMIX)	18 Gbps
IPSec VPN throughput[2] (1400B)	36 Gbps
Application security performance (TPS#/CPS**)	36 Gbps/23 Gbps
Next-generation firewall (TPS#/CPS**)[3]	35 Gbps/12 Gbps
Secure Web Access Firewall (CPS**)[4]	11 Gbps

[2] Throughput numbers based on UDP packets and RFC2544 test methodology

[3] Next-generation firewall performance is measured with firewall, application security, and IPS enabled

[4] Secure Web Access firewall performance is measured with firewall, application security, IPS, SecIntel, and URL filtering enabled

TPS Method: Throughput performance of average HTTP sessions

** CPS Method: Short-lived sessions

Specifications	SRX2300
Advanced Threat (CPS**)[5]	6 Gbps
Connections per second (64B)	450,000
SSL connections per second	8,000
Maximum concurrent sessions (IPv4 or IPv6)	5 Million
Route table size (RIB/FIB) (IPv4)	2 Million/1.2 Million
IPSec VPN tunnels	4,000

Juniper Mist WAN Assurance and AI Operations

Alternatively, the SRX2300 firewall can be operated and orchestrated through the Juniper Mist Cloud. Mist AI delivers unprecedented automation using a combination of artificial intelligence, machine learning algorithms, and data science techniques to save time, maximize IT productivity, and deliver the best experience to digital users.

Juniper Mist WAN Assurance is built on the Juniper Mist Cloud and delivers full lifecycle management and operations, including AI-Native insights, automated speed tests, dynamic packet capture (dPCAP), anomaly detection, and root cause identification that focuses on end users' experience. For Day 0 and Day 1 operations, WAN Assurance also provides orchestration, administration, and ZTP for the SRX2300. See the WAN Assurance data sheet for more information.

Ordering information

To order Juniper Networks SRX Series firewalls, and to access software licensing information, please visit the How to Buy page at juniper.net/us/en/how-to-buy/form.html.

About HPE

HPE is a leader in essential enterprise technology, bringing together the power of AI, cloud, and networking to help organizations achieve more. As pioneers of possibility, our innovation and expertise advance the way people live and work. We empower our customers across industries to optimize operational performance, transform data into foresight, and maximize their impact. Unlock your boldest ambitions, with HPE. Discover more at HPE.com.

[5] Advanced Threat performance is measured with Firewall, Application Security, IPS, SecIntel, URL Filtering and Malware Protection enabled

Visit HPE.com

Chat now

© Copyright 2025 Hewlett Packard Enterprise Development LP. The information contained herein is subject to change without notice. The only warranties Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty. Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.

Android is a registered trademark of Google LLC. Windows is either a registered trademark or trademark of Microsoft Corporation in the United States and/or other countries. All third-party marks are property of their respective owners.

a00150834ENW

HEWLETT PACKARD ENTERPRISE

hpe.com